

10 Mathematical Induction

Prove the following statements with either induction, strong induction or proof by smallest counterexample.

2. For every integer $n \in \mathbb{N}$, it follows that $1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 1$, this statement is $1^2 = \frac{(1)(2)(3)}{6} = \frac{6}{6}$, which is true.
2. Assume $1^2 + 2^2 + 3^2 + 4^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$ for some integer $n = k \geq 1$.
 Observe that this implies the statement is true for $n = k + 1$.

$$\begin{aligned}
 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 &= \\
 (1^2 + 2^2 + 3^2 + \dots + k^2) + (k+1)^2 &= \\
 \frac{k(k+1)(2k+1)}{6} + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + \frac{6(k+1)^2}{6} \\
 &= \frac{k(k+1)(2k+1) + 6(k+1)(k+1)}{6} \\
 &= \frac{2k^3 + 9k^2 + 13k + 6}{6} \\
 &= \frac{(k+1)(k+2)(2k+3)}{6} \\
 &= \frac{(k+1)([k+1]+1)(2[k+1]+1)}{6}
 \end{aligned}$$

Therefore, $1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{(k+1)([k+1]+1)(2[k+1]+1)}{6}$, which means the statement is true for $n = k + 1$. $\square$